What It Took to Read the Law:
Access, Provenance and the Evidentiary Basis of a Digital Legal Corpus

Abstract

Governments publish their law online, and a growing amount of public and private activity is built directly on those publications: compliance systems, regulatory technology, legal research, and language models grounded in statutory text. All of it assumes that the official record is available when needed and internally consistent when read. That assumption is rarely tested, because testing it requires a contemporaneous record of what each attempt to retrieve the official text actually returned. This article reports one. In building a corpus of Saudi Arabian legislation, every verified article was annotated with the sources consulted and how they were reconciled, and every instrument was assigned a verification tier by hand. Measured over the 11,704 articles whose record names its sources, the official source was unreachable for 20.0 per cent of them, covering 28.0 per cent of instruments; 20.9 per cent were obtained through a web archive, and two thirds of those cases coincide with a recorded unavailability of the live source; 8.2 per cent required optical reconstruction; and 4.9 per cent record a defect in the official text itself. Weighted by article, 17.0 per cent of the corpus has no cross-verified official primary source behind it, and the coverage of the sample makes that an underestimate. In four instruments the official record contradicts itself across nine articles: the provision the portal displays and the amendment log the same portal publishes do not agree. We draw three conclusions for the publishers and users of official data. Web archives are load-bearing infrastructure for public-record research, and hold that position with no mandate and no guarantee of permanence. Availability and internal consistency are distinct properties and need separate measurement. And a public dataset should carry the strength of the evidence behind each record, for which we propose a minimal five-field provenance schema derived from what this corpus recorded in free text — and from the 24 per cent of its own provenance values that turned out not to be provenance statements at all.

Keywords: public data infrastructure; provenance; data quality; web archiving; official records; open government data; legal informatics; Saudi Arabia

1 Introduction

Article 1 of the Saudi Environmental Law can be read on the official legislative portal. It can also be read, on the same portal, in a form that does not match. The corpus described below records the discrepancy against that article, resolved only by going to two independent sources outside the portal and taking the reading on which they agreed.[1]

That is an uncomfortable observation to build on, and it is not the point of this article. The point is what it takes to notice. A user who consults the portal once, on a good day, gets an answer

[1] Environmental Law (Royal Decree M/165, 19/11/1441H) art 1; the provenance record for the article is published with the analysis code.

and no reason to doubt it. The discrepancy is visible only to someone who consulted the portal repeatedly, kept a record of what each attempt returned, and compared. Almost nobody does that, because almost nobody has reason to: the official source is where the law is, and if the official source says something, that is what the law says.

The reason this matters beyond one statute is that the population of users has changed. When the audience for published law was lawyers reading one provision at a time, an occasional outage was an inconvenience and an inconsistency was something a professional would notice and query. The audience now includes systems: compliance engines, regulatory-technology products, public bodies consuming each other's published rules, and models grounded in statutory text. These consume the record at volume, without a professional reader between the source and the answer, and they cannot query anything. For them an outage is not an inconvenience — it is a gap silently filled from somewhere else, or a record that never enters the collection at all, with nothing downstream marking either event.

This article examines what happens to that assumption when it is checked. It uses a corpus of Saudi Arabian legislation in which every verified article carries a provenance string naming the sources consulted and how they were reconciled, together with a verification tier assigned by hand to each instrument.[2] The corpus was not built to study its own provenance. The record exists because building it required deciding, article by article, whether the text in hand could be relied on — and the record of those decisions turns out to be evidence about something larger than one corpus.

Three findings follow, and a fourth observation about this corpus's own shortcomings.

First, **the official source was frequently not available**. For 20.0 per cent of articles, covering 28.0 per cent of instruments, the build record documents that the primary official source could not be reached: a portal returning an error, a connection reset, a page that no longer resolved. This is not a claim that the sources are badly maintained. It is a measurement of how often a researcher who needs the official text will not get it on the first attempt — or the fifth.

Second, **web archives did much of the work instead**. 20.9 per cent of articles were obtained through an archived capture rather than a live official page. The two totals are close, but that alone would prove nothing — so the joint is measured rather than inferred: 67.8 per cent of the articles whose official source was unreachable were recovered through an archive, and 65.1 per cent of archive use coincided with a recorded unavailability. The archive standing in for a page that would not load is therefore the dominant pattern and not the only one; 7.3 per cent of articles used an archive with no unavailability recorded, and 6.4 per cent record unavailability that an archive did not resolve. The Internet Archive is not a fallback for this kind of research. It is part of the infrastructure, and it holds that position without any legal status, any guarantee of permanence, and no obligation to anyone.

Third, **the evidence behind the text is uneven, and nothing published alongside the text says so**. Weighted by article, 64.2 per cent of the corpus rests on two or more official sources in agreement; 18.9 per cent on one official source cross-checked against secondary ones; 4.5 per cent on secondary sources only, because no official source could be reached; and 12.5 per cent on a single source or on evidence that varies within the instrument. A reader given the consolidated text alone cannot distinguish the first case from the last.

[2] The corpus and the scripts that reproduce every number and figure here are openly archived; see the data availability statement.

The fourth observation is about the record itself, and it cuts against the corpus this article uses. The provenance field is free text, not a controlled vocabulary, and 3,704 of 15,408 articles carry a value in it that is not a provenance statement at all — a lifecycle status such as 'unchanged' that leaked into the same field. A project that took provenance unusually seriously still ended up with a provenance field that means three different things. That is reported here rather than quietly cleaned, because it is the strongest available argument that provenance needs a vocabulary and a schema rather than a place to write a sentence.

2 Who builds on the official record

The stakes of this question are not confined to research. Publishing law online converted a reference artefact into an operational one, and a large amount of public and private activity now consumes it directly rather than through a human intermediary who might notice a problem.

Compliance and regulatory technology. Systems that determine whether an obligation applies to a firm read statutory text and act on it. Where the text is retrieved once, cached, and served thereafter, a defect at retrieval time propagates silently: nothing downstream can distinguish a provision transcribed from a stable official page from one recovered from an archived capture of a page that no longer resolves.

Government's own systems. Public bodies consume each other's published rules. An agency implementing a licensing regime relies on the published text of the law that creates it, and on the published text of every instrument that one cross-references. Section 6 describes four instruments where the official record disagrees with itself; a public body reading the portal has exactly the same exposure as a private one.

Language models grounded in statutory text. Retrieval-augmented systems answer from a corpus, and the corpus was collected at a moment. The literature on these systems concentrates on retrieval quality and on hallucination, both of which presume that the retrieved text is the law. What this article measures is the layer below: whether the text entered the corpus correctly in the first place, and how strongly it is attested.

Open government data more generally. Legal publication is one instance of a wider pattern. States publish registries, budgets, procurement notices, statistical series and legal texts, and the same three questions apply to all of them: was it available when it was needed, is it internally consistent, and does anything published alongside the data say how strongly it is attested? Law is a useful place to ask because the correct answer is unusually well defined — there is a right text, published in a gazette, and divergence from it is an error rather than a matter of interpretation.

2.1 In citation practice

Legal citation is built on the premise that a citation is a retrieval instruction: it names an instrument and a provision so that a reader can go and read it. The convention is old enough to predate the question of whether the naming is enough, and it does not encode where the reader should go, or what they should do if what they find there is not what the citing author found.

For most of the history of legal citation this was adequate, because the authoritative artefact was a printed gazette held in libraries, and the question 'is this the text?' had a physical answer. It is less adequate when the authoritative artefact is a database behind a web front end, whose contents can change without notice and whose availability is a function of infrastructure nobody involved in the citation controls.

2.2 In empirical legal research and legal AI

The exposure is sharper for work that consumes law at scale. A study measuring the structure of a legal system, or a model trained or grounded on statutory text, takes a snapshot and treats it as the law. Machine-readable legal corpora have grown substantially in recent years, and their documentation concentrates on size, licensing and composition rather than on how the text was obtained or how strongly it is attested.[3] The dataset-documentation literature has argued for exactly this kind of disclosure in general terms, and legal corpora have largely not adopted it.[4]

Structural standards for legislative documents can carry provenance metadata, and the capacity has been available for years.[5] The gap is not representational. It is that nobody is required to fill the fields, and a corpus that leaves them empty looks exactly like a corpus whose text is impeccable.

2.3 In the data-stewardship literature

The general case for recording provenance is settled. The FAIR principles make 'Reusable' conditional on data being 'associated with detailed provenance', and they are endorsed across research funders, publishers and infrastructures.[6] A standard vocabulary for expressing it has existed since 2013.[7]

What this article adds is not another argument that provenance matters. It is an account of what provenance turns out to be *for*, once a collection is large enough that nobody can remember how any particular record was obtained. The FAIR formulation — 'detailed provenance' — is satisfied, on its face, by the free-text field this corpus used. That field was written conscientiously, article by article, by people who cared about the question, and it still could not be counted without first classifying 158 distinct strings by rule, and still contained 24 per cent of values that were not provenance at all. 'Detailed' is not the operative property. *Closed* is: a small set of fields with enumerated values, which is what section 7 proposes.

[3] Peter Henderson and others, 'Pile of Law: Learning Responsible Data Filtering from the Law and a 256GB Open-Source Legal Dataset' (NeurIPS 2022, Datasets and Benchmarks Track); Joel Niklaus and others, 'MultiLegalPile: A 689GB Multilingual Legal Corpus' (Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics, 2024).

[4] Timnit Gebru and others, 'Datasheets for Datasets' (2021) 64 Communications of the ACM 86; Emily M Bender and Batya Friedman, 'Data Statements for Natural Language Processing' (2018) 6 Transactions of the Association for Computational Linguistics 587.

[5] Monica Palmirani and Fabio Vitali, 'Akoma-Ntoso for Legal Documents' in *Legislative XML for the Semantic Web* (Springer 2011) 75.

[6] Mark D Wilkinson and others, 'The FAIR Guiding Principles for Scientific Data Management and Stewardship' (2016) 3 Scientific Data 160018.

[7] W3C, *PROV-O: The PROV Ontology* (W3C Recommendation, 30 April 2013).

2.4 In the archiving literature

That legal citations rot is not a new observation. Zittrain, Albert and Lessig found that more than 70 per cent of URLs cited in the *Harvard Law Review* and comparable journals, and 50 per cent of those in United States Supreme Court opinions, no longer led to the material originally cited — the finding that produced Perma.cc.[8] What that literature measures is the failure of a citation to resolve some time after publication. This article measures something adjacent and, for a corpus builder, more immediate: the failure of an official source to resolve *at the moment of collection*, when the researcher is actively trying and has every incentive to succeed.

3 Data and method

3.1 The provenance record

Each verified article in the corpus carries a status string written during the build, naming the sources consulted and how they were reconciled — for example, that the Ministry of Justice portal's API text was cross-checked against the official PDF from the same portal, or that the live official portal was unreachable and the text rests on two independent secondary sources agreeing word for word. There are 15,408 such articles and 158 distinct strings.

Two examples convey the register better than a description. One article's status reads, in effect, that the Ministry of Justice portal's API text was cross-checked against the official PDF from the same portal — a single official channel, two artefacts, no difficulty. Another records that the live legislative portal was unreachable, that the text was taken from an archived capture of that portal, and that it was cross-verified against two independent reproductions and the official gazette, with one ligature defect from PDF extraction corrected and disclosed. Both are one field of one record. The first is four tokens long; the second is thirty.

Each instrument also carries a verification tier assigned by hand: two or more official sources in agreement (tier 1); one official source cross-checked against secondary sources (tier 2); no official source reachable, secondary sources only (tier 3); or a single source, or evidence that varies within the instrument (tier 4).

3.2 Classifying free text, and publishing the classification

The status strings are not a controlled vocabulary, and treating them as one would be the first mistake available. The analysis therefore classifies each distinct string by an explicit rule and publishes the entire mapping, so that a reader can disagree with any single assignment rather than having to accept the totals.

A string counts as a *provenance statement* when it names a source, an artefact or a retrieval route — a portal, a gazette, a named secondary site, a PDF, an archive capture, an OCR pass. Strings that name none of these are not provenance statements: they are lifecycle values such as 'unchanged' or 'amended', and in one case a single Arabic word, that were written into the same

[8] Jonathan Zittrain, Kendra Albert and Lawrence Lessig, 'Perma: Scoping and Addressing the Problem of Link and Reference Rot in Legal Citations' (2014) 127 Harvard Law Review Forum 176.

	Articles	Instruments
Official source unreachable	20.0%	28.0%
Reached through a web archive	20.9%	28.0%
Single source only	10.2%	10.0%
Required optical or visual reconstruction	8.2%	14.0%
Defect recorded in the official source	4.9%	6.5%
Multi-source cross-verification	86.1%	87.5%

Table 1: Measured over the 11,704 articles whose status field is a provenance statement, across 200 instruments.

field. There are 3,704 such articles, 24.0 per cent of those carrying any status at all. They are excluded from every measure below.

Within the provenance statements, six measures are computed by token match, each reported as a share of articles and as a share of instruments. The measures overlap by design: an instrument reached through an archive because the live portal was down scores on both. Every count is published with its underlying string table.

3.3 Two decisions that changed the numbers

Two methodological choices are recorded because each moved a headline, and a reader is entitled to know which way.

Weighting. Measures are computed per article, not per instrument. A single-source instrument of five articles and one of three hundred are not the same exposure, and an instrument count cannot tell them apart. The two denominators diverge in ways that are themselves informative: optical reconstruction touches 8.2 per cent of articles but 14.0 per cent of instruments, which is what it looks like when the hardest sources to obtain are also the shortest documents.

A prior estimate was wrong. An earlier pass over the registry's free-text notes, matching patterns against prose rather than against the structured provenance field, suggested that 47 per cent of instruments recorded an unreachable official source. The disciplined measure is 28.0 per cent of instruments and 20.0 per cent of articles. The first figure was an artefact of matching loosely-worded notes, and it is reported here because a reader encountering both should know which to use and why.

4 What it took to obtain the text

Table 1 and Figure 1 give the result. Four things in it are worth drawing out.

The last row is the good news and it should be read first. 86.1 per cent of articles were cross-verified against more than one source. This is not a portrait of a corpus assembled carelessly from whatever was to hand, and the other rows should be read in that light: they describe what a careful process had to do, not what a careless one got away with.

Unavailability and archive use are close, and they overlap by about two thirds. The totals are 20.0 per cent against 20.9 per cent, with 28.0 per cent of instruments on both — but matching totals do not establish that the same articles are involved, so the joint distribution is computed. Of the articles whose official source was unreachable, 67.8 per cent were recovered through an archive;

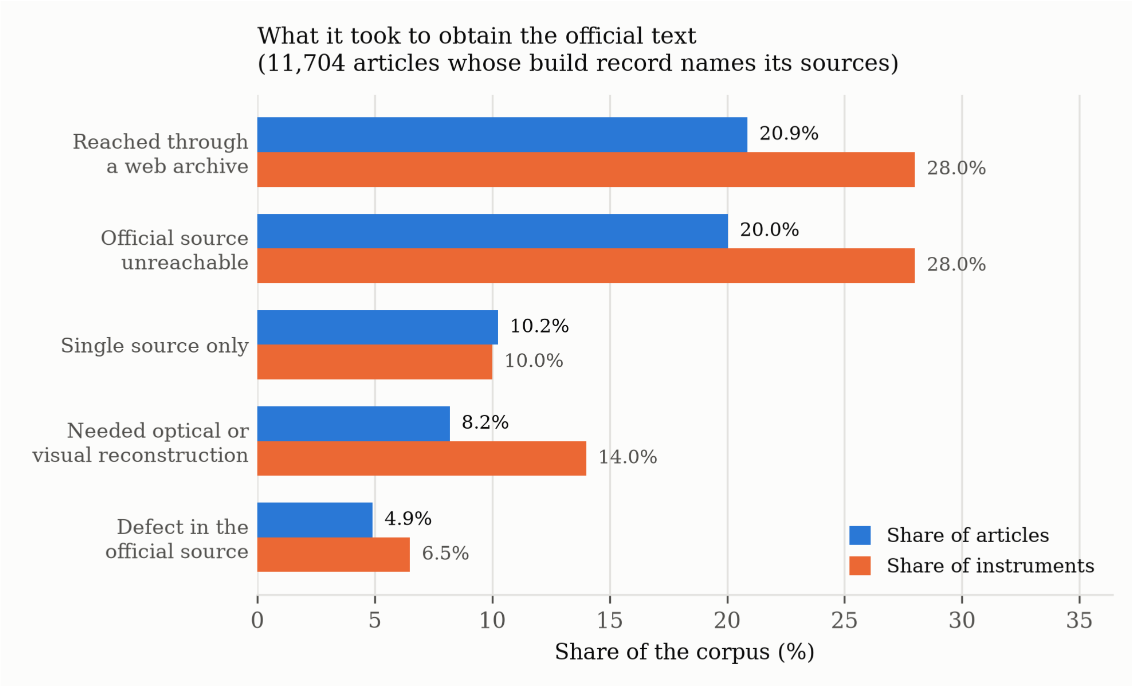

Figure 1: The friction documented in the build record, as a share of articles and of instruments. The measures overlap: an instrument reached through an archive because the live portal was down appears in both of the top two rows.

of the articles obtained through an archive, 65.1 per cent had a recorded unavailability. The archive is mostly not supplying different or better material: it is supplying the official material, at a moment when the official channel could not. The remainders matter too, and neither is small: 852 articles used an archive with no unavailability recorded — an archived capture preferred for its stability, or consulted to check whether a live page had changed — and 754 record an unavailability that no archive resolved.

Optical reconstruction is an instrument-level burden. 8.2 per cent of articles but 14.0 per cent of instruments. The sources that had to be read as images — scanned resolutions, PDFs with broken text layers, ministry documents published as pictures of pages — are disproportionately short instruments. One in seven instruments in this corpus could not be read by machine at all without an optical pass, and each such pass is an opportunity for a silent error that no downstream user can detect.

Defects in the official text are rarer than access failures but not rare. 4.9 per cent of articles carry a record of a defect in the source itself: a typographical error preserved rather than silently corrected, ligature corruption from PDF extraction, a scrambled text layer, a truncated page. These are recorded because the build policy was to preserve source defects verbatim and document them, rather than to tidy them. A corpus that tidies them looks cleaner and tells its users less.

5 The strength of the evidence

Figure 2 weights the hand-assigned tiers by article. Just under two thirds of the corpus rests on two or more official sources in agreement. The remaining third does not, and 17.0 per cent has

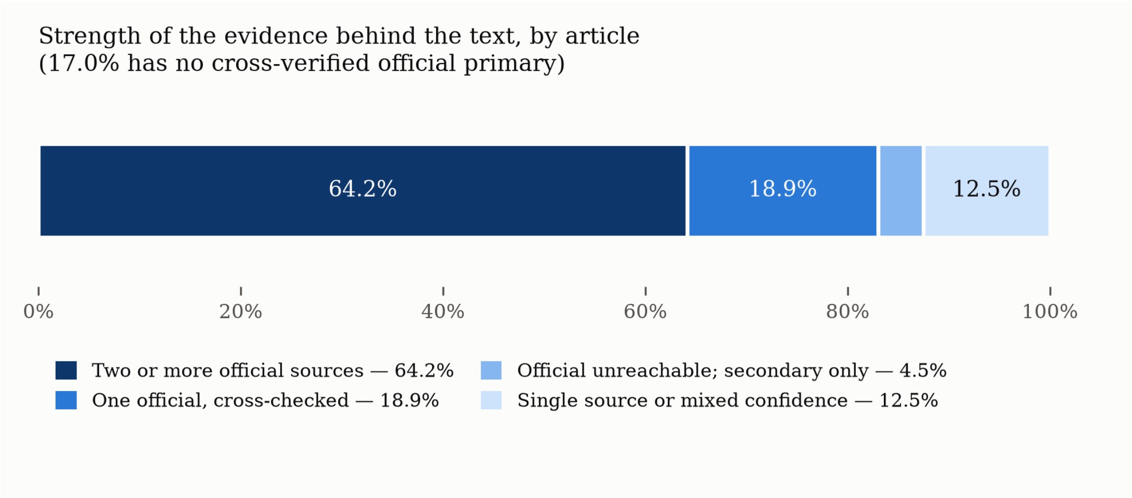

Figure 2: Verification tiers weighted by article. The tiers are ordinal, from two or more agreeing official sources to a single source or evidence that varies within the instrument.

no cross-verified official primary source at all: 4.5 per cent because no official source could be reached, and 12.5 per cent because the evidence is single-source or varies within the instrument.

5.1 What a tier means in practice

The tiers are easier to trust with two instruments behind them.

At the top, the Regulation Organizing the Work of Judicial Assistants: every article was taken from the Ministry of Justice portal's database and checked, article by article, against the official PDF published by the same ministry from the same portal. Two official artefacts, no access failure, agreement throughout. A user who needs to be certain of a provision in that instrument can be about as certain as the published record permits.

At the bottom, the Accredited Valuers Law — and it illustrates something more useful than a simple contrast. Forty of its forty-five articles were taken from the administering authority's own official PDF and cross-verified against the official gazette and two independent reproductions, which is close to the strongest evidence in the corpus. The remaining five rest on a single source, because the primary legislative portal could not be reached for them. The whole instrument is graded tier 4.

That is the scheme working as designed: a tier describes the weakest evidence anywhere in the instrument, because a user asking about one provision has no way to know whether it falls in the strong forty or the weak five. It also means the tier figures are pessimistic at the instrument level, which section 5.2 takes up alongside the bias that runs the other way.

Between them sits the largest group by instrument count: one official source reached, then cross-checked against independent non-official reproductions. That is a defensible standard and it is not the same as two official sources agreeing, because the non-official reproduction may itself derive from the official one — a possibility this corpus flags where it was detected but cannot exclude in general.

5.2 The direction of the error

Two biases act on the 17.0 per cent figure, and they pull in opposite directions. Reporting only the one that flatters the argument would be the move a reader should most distrust, so both are stated.

Sample coverage understates it. Not every instrument has article-level verified records, and the ones that do are not a random draw. Coverage by tier is 87 per cent for tier 1, 59 per cent for tier 2, 28 per cent for tier 3 and 73 per cent for tier 4. The pattern is not monotonic, but tier 3 — the instruments whose official source could not be reached at all — is by a wide margin the least represented, so the instruments most likely to be missing are those whose evidence is weakest. A registry-wide count of instruments, which does not have this problem, puts tiers 3 and 4 together at 76 of 291 instruments, or 26 per cent.

Worst-part grading overstates it. A tier is assigned to an instrument on its weakest evidence, so an instrument like the Accredited Valuers Law — forty articles at close to the strongest standard and five at the weakest — contributes all forty-five of its articles to tier 4. Article-weighted tier figures therefore attribute strong evidence to a weak tier whenever the weakness is localised.

We cannot net these against each other, and we do not try. What can be said is narrower and still worth saying: the share of this corpus that a careful user would want to treat as less than fully attested is somewhere in the region of a sixth to a quarter, depending on which unit is counted and how conservatively a mixed instrument is graded — and it is not the zero that a dataset shipping text alone implicitly asserts.

6 When the official record disagrees with itself

The sharpest cases are the ones where the problem is not access but consistency. In four instruments the build record documents the official source contradicting itself, across nine articles in total.

A note on that count, because it is easy to inflate. A provenance string is attached to every article of the instrument it describes, so it records the scope of the *verification route*, not the scope of the problem. The Environmental Law's string rides on all 49 of its articles; its registry note records that 48 matched verbatim across three sources and one did not. The figure below is therefore read from each instrument's own account of its discrepancy, and the count of articles merely carrying one of these strings — 57 — is reported in the results file as the upper bound it is.

The pattern in three of the four is the same. The legislative portal displays a consolidated article, and the same portal publishes a change log recording the amendments made to it; the two do not agree, because the change log has been updated and the displayed body has not.

In the **Press Law** this affects six articles — 5, 9, 36, 37, 38 and 40 — for each of which the portal simultaneously carries a changed-article marker and a fully quoted amendment log while its main displayed body still shows the pre-amendment text.

In the **Law of Engineering Practice** it affects article 1, and the disagreement is three-way rather than two: the portal's change log quotes a Council of Ministers resolution substituting a ministry name, while the portal's own main body has shown a third, different wording at every snapshot checked since 2019 — a wording that predates the resolution.[9] The professional body's own published PDF agrees with neither, and the corpus records the case as unresolved.

[9] Council of Ministers Resolution 250, 7/4/1444H.

In the **Travel Documents Law** it affects article 6, where the portal's amendment log records an amendment and omits the decree citation for it entirely; the citation was recovered from a secondary source.

The fourth is the **Environmental Law** article with which this article opened. There the disagreement is confined to a single definition — 'the competent authority' in article 1 — where the portal's per-article amendment log gives a current wording that its own main running text does not carry. It was resolved against two independent sources outside the portal, and the corpus flags it as warranting dedicated legal review rather than treating it as settled.

Two things follow. The first is practical: a researcher relying on a single retrieval from a single official page has no way to detect any of this, because detecting it requires reading two parts of the same portal against each other and noticing that they differ. The second is that a consolidated text and its own amendment history are two claims about the law, and a system that publishes both without reconciling them has published a contradiction. Sixteen instruments in this corpus carry a documented staleness risk of exactly this kind.

7 A minimal provenance schema

The most transferable output of this exercise is not a percentage. It is the observation that everything measured above was recoverable only because someone wrote down, at the time, what each retrieval attempt returned — and that writing it in prose made the record nearly unusable at the scale where it mattered. Twenty-four per cent of the values in this corpus's provenance field are not provenance statements at all, and the remainder had to be classified by a rule over free text before anything could be counted.

What was actually needed is small. The five fields below carry everything the analysis in this article extracted from 158 distinct free-text strings, and they are derived from that record rather than proposed in the abstract: each field exists because the free text was already trying to express it.

Three properties of Table 2 matter more than its particular field names.

It is **cheap**. Four of the five fields are closed vocabularies of three or four values, and a collection process that knows how it obtained a document knows all of them at the moment of collection. The expensive part of provenance is reconstructing it afterwards, which is what this article had to do.

It **separates availability from consistency**. `retrieval_route` records whether the live source worked; `discrepancy` records whether the sources agreed. These are different failures with different remedies, and a single confidence score — the usual shorthand — collapses them. The corpus studied here has instruments that were perfectly available and internally contradictory, and instruments that were unreachable but, once recovered, in complete agreement across three sources.

It **degrades honestly**. A record with `corroboration = 0` and `transformation = optical-recognition` is not hidden by an aggregate; a user filtering for reliability can exclude it, and a user who does not at least cannot claim they were not told. The verification tiers this corpus assigned by hand are a coarser version of the same idea, and even four levels turned out to be more than almost any comparable dataset publishes.

Field	Values and purpose
`source_class`	*official-primary* (the publishing authority's own channel), *official-secondary* (another public body republishing it), *independent*, *aggregator*. Answers: whose copy is this?
`retrieval_route`	*live*, *archived-capture*, *offline-artefact* (a PDF or scan held locally), *proxy*. Answers: how was it obtained, and would the same request work today?
`corroboration`	Integer: the number of independent sources found to agree. Zero means single-source. Answers: how strongly is it attested?
`transformation`	*none*, *text-extraction*, *optical-recognition*, *manual-transcription*. Answers: what could have introduced an error that no reader can see?
`discrepancy`	Null, or a short structured note: which sources disagreed, and whether the disagreement was resolved. Answers: is there a known problem with this record?

Table 2: A minimal provenance schema, derived from what the free-text record in this corpus was already trying to say.

8 What follows

Cite the retrieval, not just the provision. If the official source resolves four times out of five at best, then a citation that names only the instrument and the article is an incomplete instruction. What is missing is cheap to add: the date of retrieval and, where the live source failed, the archived capture actually used. Legal citation systems already accommodate 'accessed' dates for online material; the argument here is that statutory citation needs them too, for the same reason and with better evidence.

Ship the strength of the evidence with the text. The tier scheme used here is crude — four levels, assigned by hand — and it is more than almost any legal corpus publishes. A reader of a corpus that ships text alone must assume uniform reliability, and this corpus demonstrates that the assumption is false within a single collection: the same file format, the same schema, the same apparent authority, and evidence ranging from two agreeing official sources to a single scan read by machine. The cost of publishing a tier is one field.

Treat the archive as infrastructure, and fund it accordingly. One article in five in this corpus reached it through a web archive. That dependency is invisible in the finished product and total in its effect: had the captures not existed, those texts could not have been verified at the time they were collected, and in some cases could not have been obtained at all. Web archives occupy this position for legal research without any mandate to do so and without any legal recognition of the role. Whether they should is a question for the institutions that publish law, not for the archives.

Publish availability as a service property, not a hope. A state that publishes its law online has taken on an operational commitment whether or not it has said so. The measurements here are not a service-level study and cannot be one — they record what one collector saw from one network over one period — but they are enough to show that the commitment is not currently being met at the level users assume. The remedy is ordinary infrastructure practice: publish uptime, publish a stable retrieval interface, and treat a consolidated text and its own amendment log as a single artefact that must agree before either is served.

Deposit, rather than depend on someone else archiving. One article in five in this corpus reached it through a web archive. That is a dependency on a third party with no obligation to the state whose law it is preserving, and it is invisible to everyone downstream. States that already operate legal deposit for print have the institutional machinery for the digital equivalent; what is missing is the instruction to use it for the state's own published record, and a citable identifier for each version so that a reader can retrieve what an author actually read.

Use a vocabulary, not a sentence. The finding that indicts this corpus is the one most useful to the next builder. Free-text provenance decays: it accumulates conventions, absorbs values that belong in other fields, and becomes unanalysable at precisely the scale where analysis matters. Nearly a quarter of the status values here are not provenance statements at all. A short controlled vocabulary — source class, retrieval route, cross-verification count, defect flag — would have carried the same information, and this article's analysis would have taken a tenth of the effort and made none of the judgement calls that section 3 had to disclose.

8.1 Why this is not already standard

If the schema is as cheap as section 7 claims, its absence needs explaining. Three reasons seem to us to account for most of it, and none is about cost.

Provenance is invisible when everything works. A collection assembled in a week from a responsive source has nothing interesting to record, and the value of having recorded it appears only later — when a source changes, a discrepancy surfaces, or someone asks how a particular record was obtained. By then the information is gone. This is the ordinary structure of an under-provided good, and it is why the field has to be populated by the collection process rather than by whoever eventually needs it.

Publishing weakness looks like admitting weakness. A dataset that marks a tenth of its records as single-source invites the question why, and a dataset that marks none invites nothing. The incentive runs the wrong way, and it runs that way for public bodies most of all. The counter-argument is that the weakness exists whether or not it is marked, and that the unmarked version transfers the risk to users who cannot see it — but that argument has to be made, because the default is silence.

There is no convention to comply with. Dataset documentation practice addresses composition, collection and intended use; provenance at the level of the individual record is not part of what a dataset is normally expected to carry. A convention would change this faster than any argument about incentives, which is the practical case for proposing a small and boring one rather than a complete ontology that already exists and is not used.

8.2 What a user can do in the meantime

Most consumers of official data cannot change how it is published, and the recommendations above are addressed to people who can. Three things are available to everyone else, and this corpus is evidence for each.

Retrieve more than once, and keep what you got. Every discrepancy reported in section 6 was found by comparing retrievals — across time, across two parts of the same portal, or against an independent reproduction. A single retrieval cannot detect any of them. Keeping the response, rather than only the parsed result, costs almost nothing and is the difference between noticing a source change and being silently overtaken by it.

Treat a second source as a test, not a backup. The value of the independent reproductions in this corpus was rarely that they supplied text the official source could not. It was that they disagreed on 4.9 per cent of articles and thereby located the defects. A pipeline that consults a second source only when the first fails gets the resilience and forgoes the measurement.

Record what you could not do. The most useful entries in this corpus's provenance field are the ones admitting a shortfall: single-source, portal unreachable, amendment text unobtained, three-way divergence unresolved. Recording a gap costs a field and preserves the option of returning to it; leaving it blank makes a weak record indistinguishable from a strong one, including to the person who created it.

9 Applying this elsewhere

Nothing in the method is specific to law, to Saudi Arabia, or to this corpus. It requires one thing that is rare and not difficult: that whoever collects a public dataset records what each retrieval attempt returned, at the time, in a form a machine can read.

A group building any collection from an official source can produce the measurements in this article by populating Table 2 during collection and running three counts afterwards — the share of records whose live source failed, the share recovered through an archive and the overlap between them, and the share resting on a single source. The analysis here is about two hundred lines of code, and most of that is the classification of free text that a schema would have made unnecessary.

Two adjustments would be needed outside the legal domain. Where a dataset has no single authoritative version — a statistical series revised on a schedule, say — `discrepancy` carries more weight than `corroboration`, because disagreement between two sources may be a legitimate revision rather than an error. And where records are produced continuously rather than enacted, availability should be sampled over time rather than observed once at collection, which turns a build-time snapshot into the monitoring study this one explicitly is not.

The comparison that would be most useful is the obvious one: the same measurements, for the same kind of source, in several jurisdictions. This article can say that a fifth of one state's published legal record was unreachable when someone tried to collect it. Whether that is typical, unusually good or unusually bad is not knowable from a single case, and it is the first question any reader should ask.

10 Limitations

One corpus, one jurisdiction, one build. Everything here describes the sources of Saudi legislation as encountered during a particular construction period. Availability is a moving target; a portal unreachable then may be reliable now, and the reverse. What generalises is the method and the observation that the question is worth asking, not the specific percentages.

The record is the builder's own. The provenance strings were written by the same project whose reliability they describe. There is no independent audit, and a record of this kind cannot show what it failed to notice. Its usefulness rests on having been written contemporaneously, article by article, and on being published in full for inspection.

Classification is by rule over free text. Section 3 sets out the rule and the full string-to-classification table is published. It is nonetheless a rule applied to sentences, and a different rule would give somewhat different numbers. The direction of that uncertainty is not knowable in advance, unlike the sample-coverage bias, which is.

'Unreachable' is a build-time observation, not a service-level measurement. The records document failures encountered by one collector, from one network, over one period. They are not a monitoring study, and they cannot distinguish an outage from a block, a rate limit, or a route that happened to fail from where the request originated.

Article-level coverage is uneven. As section 5.1 sets out, the weakest tier is the least represented in the article-weighted measures. The instrument-weighted figures do not have this problem and are reported alongside throughout.

11 Conclusion

Governments publish law, and increasingly the audience is machinery. This article asked whether the record that machinery consumes is available when needed and consistent when read, using the only evidence that can answer it: a contemporaneous account of what each retrieval attempt actually returned.

For roughly two thirds of one state's published legal corpus, the answer is straightforwardly yes — two or more official sources, in agreement, obtained without difficulty. For the rest it is qualified. The official source could not be reached for one article in five. One article in five arrived through a web archive, and two thirds of those cases coincide with a live source that would not respond. One instrument in seven had to be read as images. Around a sixth to a quarter of the corpus, depending on the unit counted, rests on evidence a careful user would want to treat as less than fully attested. In four instruments the record contradicts itself, and in each the contradiction is between two things the same official portal publishes.

None of this makes the law unknowable, and none of it is an accusation. Legal publication in this jurisdiction is extensive, it is free, and for most of the corpus it worked. The finding is narrower and more actionable: the gap is between how those who build on official data assume it behaves and how it was observed to behave across one sustained attempt to use it — and nothing published alongside the data tells a user which case they are in.

Three things would close that gap, and none requires better sources, which is nobody's to promise. Record availability as a service property rather than an assumption. Deposit the record rather than relying on third-party archives that have taken on no obligation. And carry the strength of the evidence with each record, in five closed fields rather than a sentence.

The last of these is the one this article can demonstrate rather than argue. Everything measured here was recoverable only because someone wrote down, at the time, what each attempt returned. That they wrote it in prose is why the analysis needed a classifier, a published mapping and two rounds of correction before it could report a number — and why nearly a quarter of the values turned out not to be provenance at all. The record was worth keeping. It was kept in the wrong shape. Both halves of that sentence are the finding.